Task result overview

Task information

Software version: 1.5

Generated UTC: 2026-10-12T05:46:07Z

Task name: sample_report_120s

Analysis duration: 0.03333 h

Time step: 60 s

Thermal network: reference_11_node_gnss_receiver

Outputs from this run

scenario

Main findings

Mean beta angle: 19.46 deg

Eclipse fraction: 0 %

Response-node temperature P95-P5: 0.3088 K

Simulated code-bias P95-P5: 0.108 m

Power-to-temperature peak time lag not calculated: no cycle covers at least 90% of an orbit with at least 8 samples.

Maximum energy closure: 4.475e-10 J

Orbit heating and nodal response

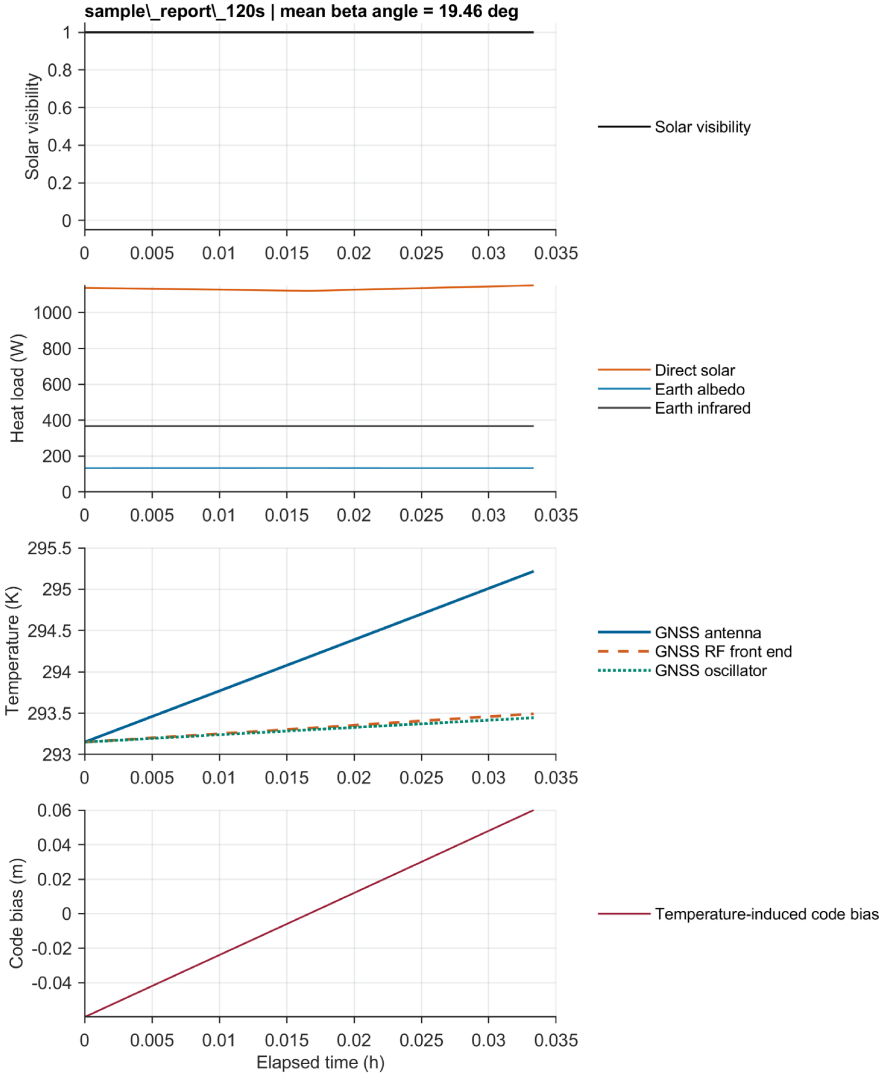
Mean beta angle: 19.46 deg; eclipse fraction: 0.00%.

Response-node temperature P95-P5: 0.3088 K; assumed temperature-induced code-bias P95-P5: 0.108 m.

Power-to-temperature peak time lag not calculated: no cycle covers at least 90% of an orbit with at least 8 samples.

With almost no eclipse, post-eclipse thermal lag cannot be assessed.

No warmup was used; initial-temperature transients may affect spans and lag.



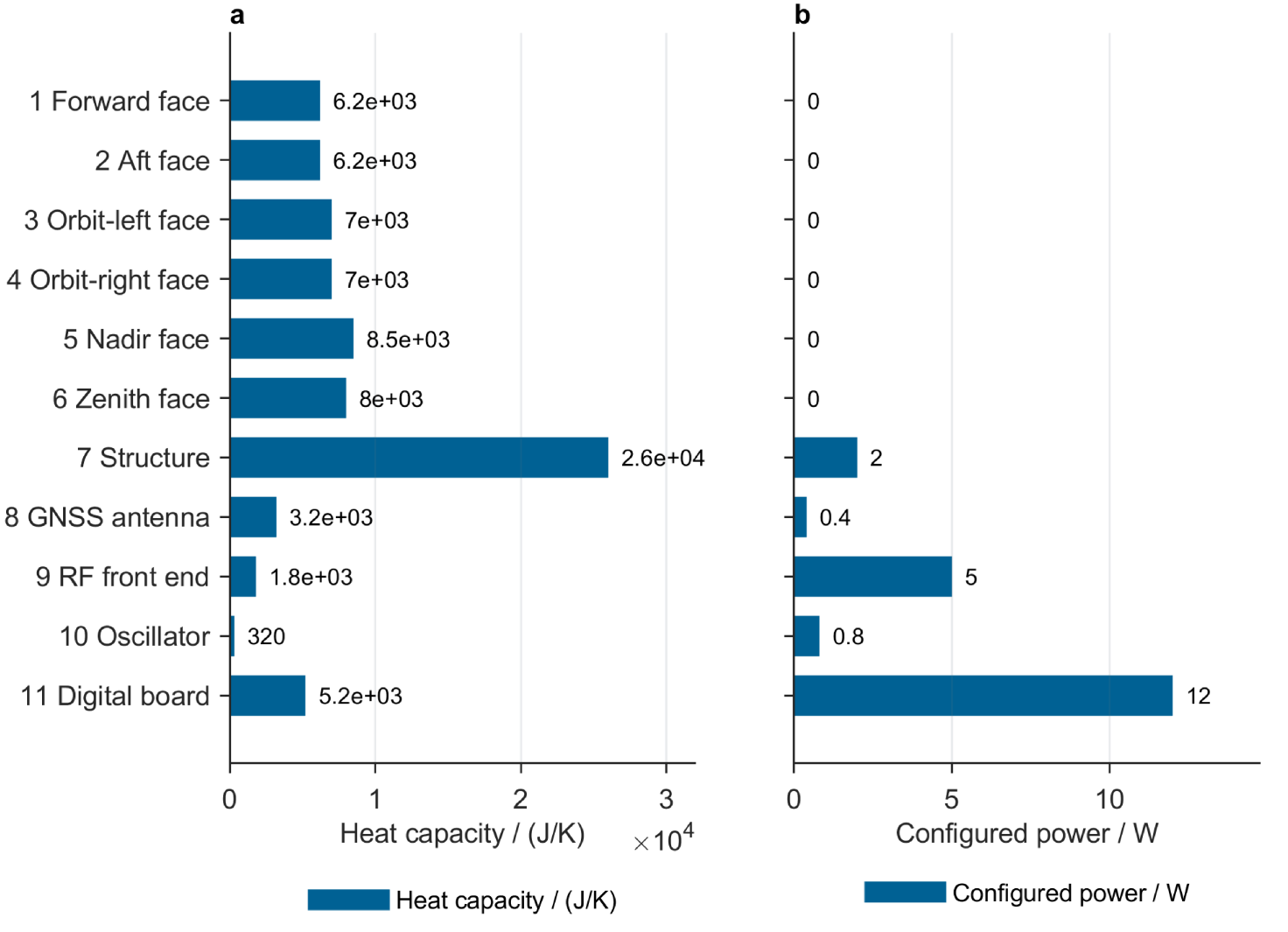
Heat capacity and internal dissipation

11 nodes; total capacity 79420 J/K; configured internal power 20.2 W.

Spacecraft structure has the largest capacity: 26000 J/K (32.7% of total).

Receiver digital board has the highest configured power: 12 W.

11/11 nodes plotted, selected by full temperature range; all are tabulated.



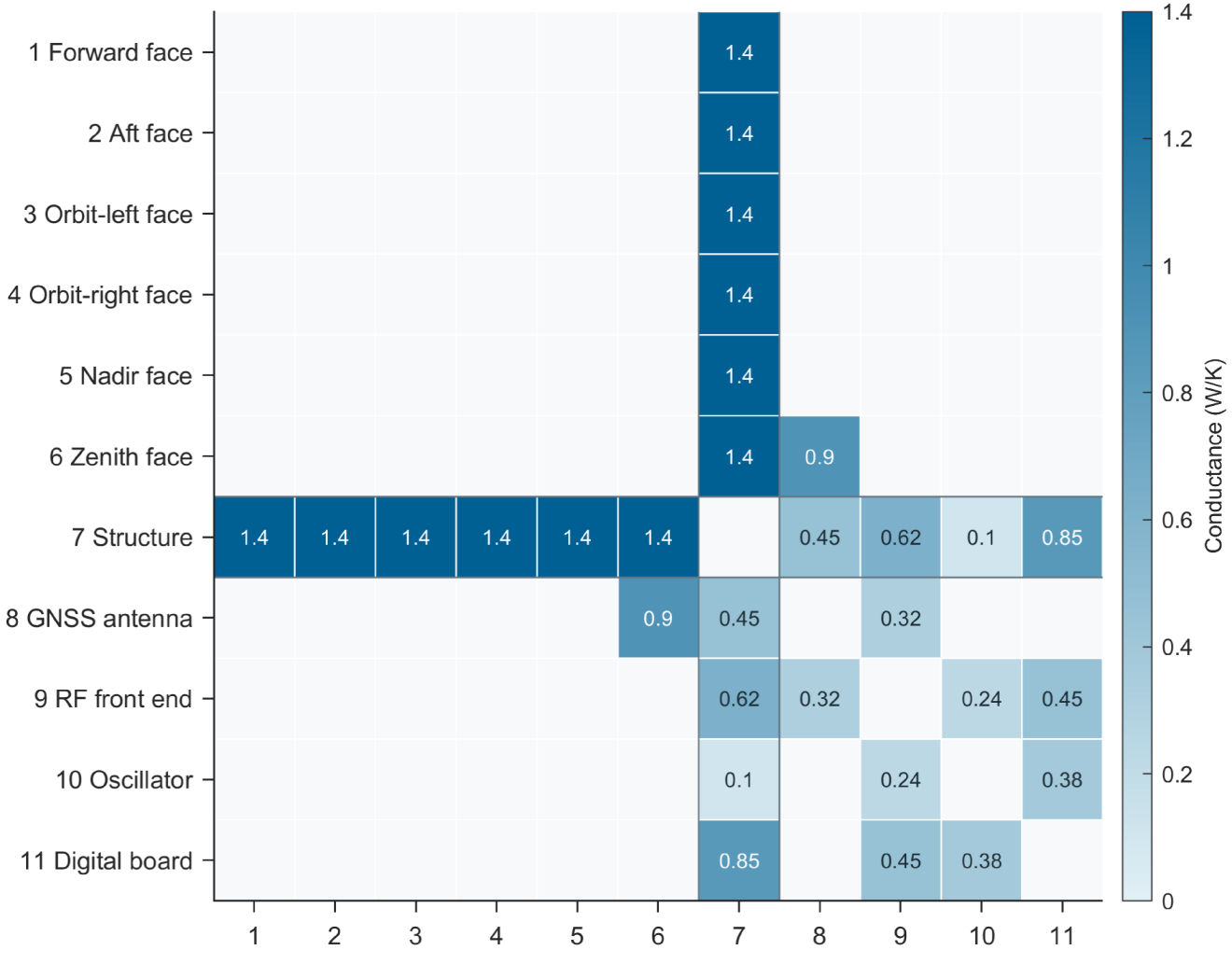
Thermal links and conductance

15 conductive links; Spacecraft structure has the most connections (10).

First 11/11 nodes shown; filled cells and values give conductance. Blank cells have no direct conductive link.

The symmetric matrix shows each reciprocal link twice. Conductances are listed in the link CSV.

Among links connected to GNSS RF front end, the largest absolute mean net transfer is GNSS antenna to GNSS RF front end at 0.276 W. Total mean net conduction from the response node is -0.146 W (positive outward). This identifies an active path using heat flow rather than link count.

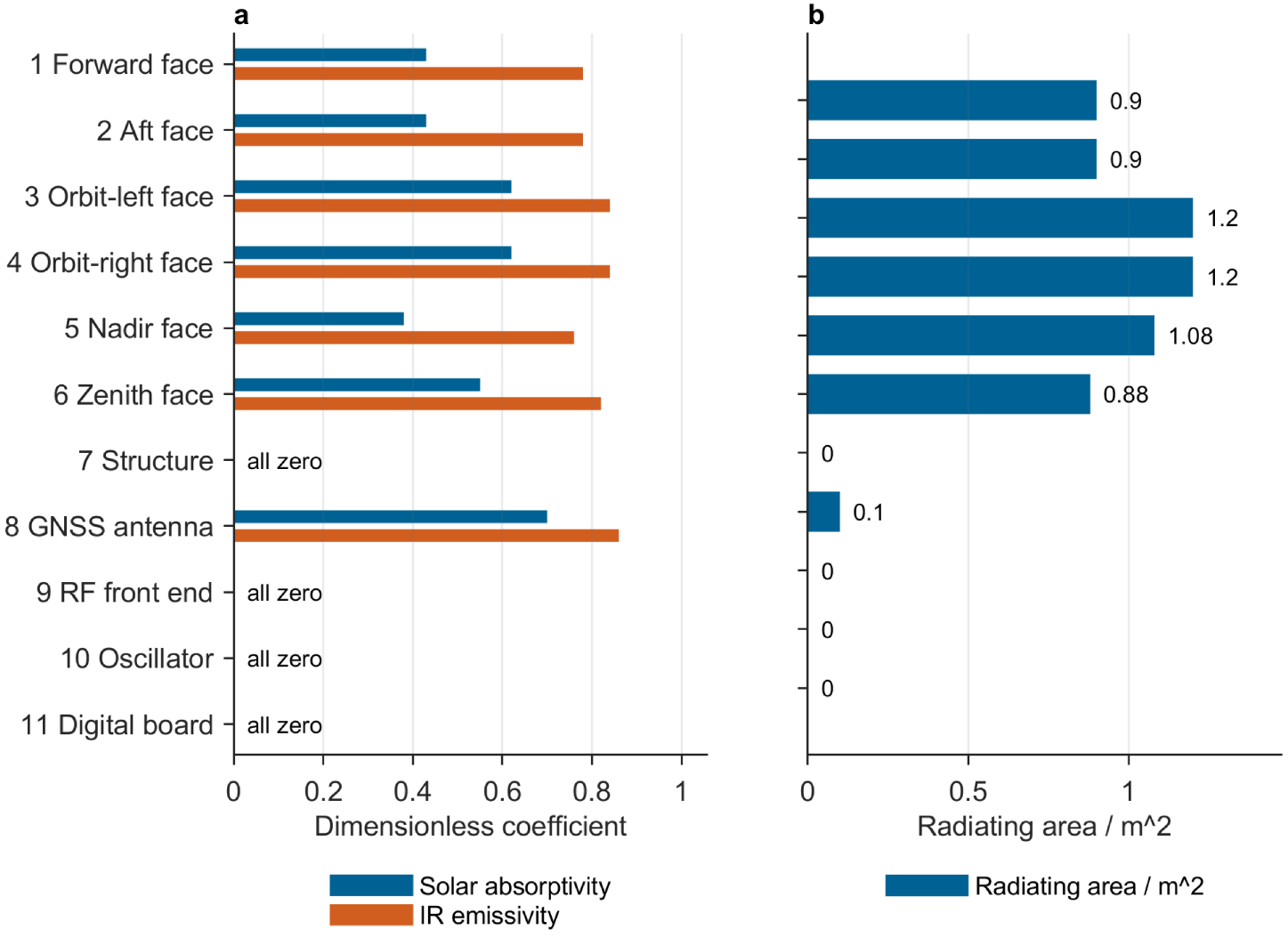


Optical properties and radiating area

Total external radiating area: 6.26 m^2. Zero values are explicitly labelled.

11/11 nodes plotted, selected by full temperature range; all are tabulated.

Spacecraft structure, GNSS RF front end, GNSS oscillator, Receiver digital board have zero projected and external radiating areas and exchange heat through conductive links.



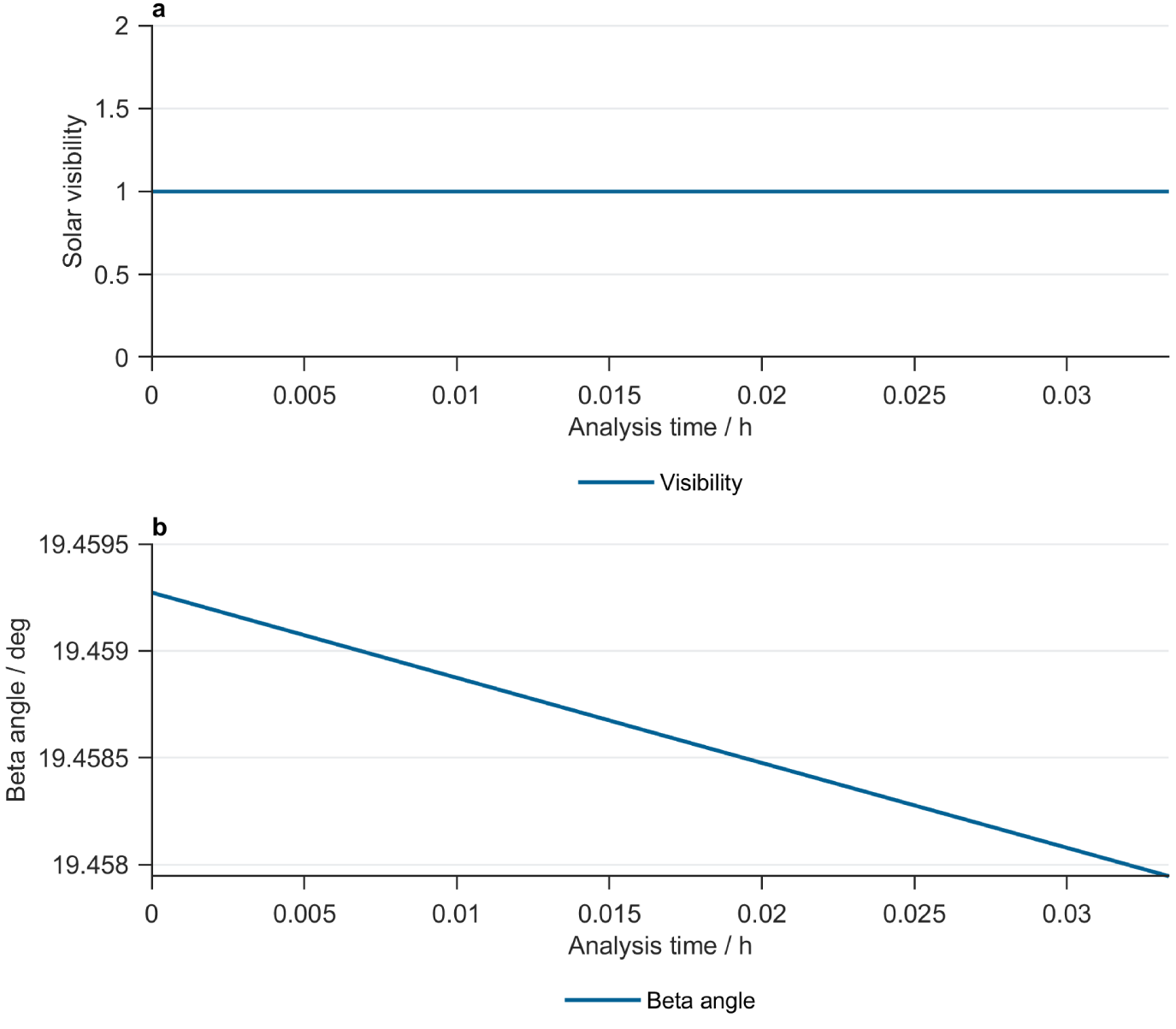
Illumination and eclipse structure

Trapezoidal time weighting estimates partial/full eclipse over 0.00% of the window.

Solar visibility ranges from 1.000 to 1.000; orbital period 96.69 min.

Eclipse uses visibility below 0.999. Time-weighted and epoch-count fractions can differ.

0 eclipse entries and 0 exits; beta angle 19.458 to 19.459 deg.



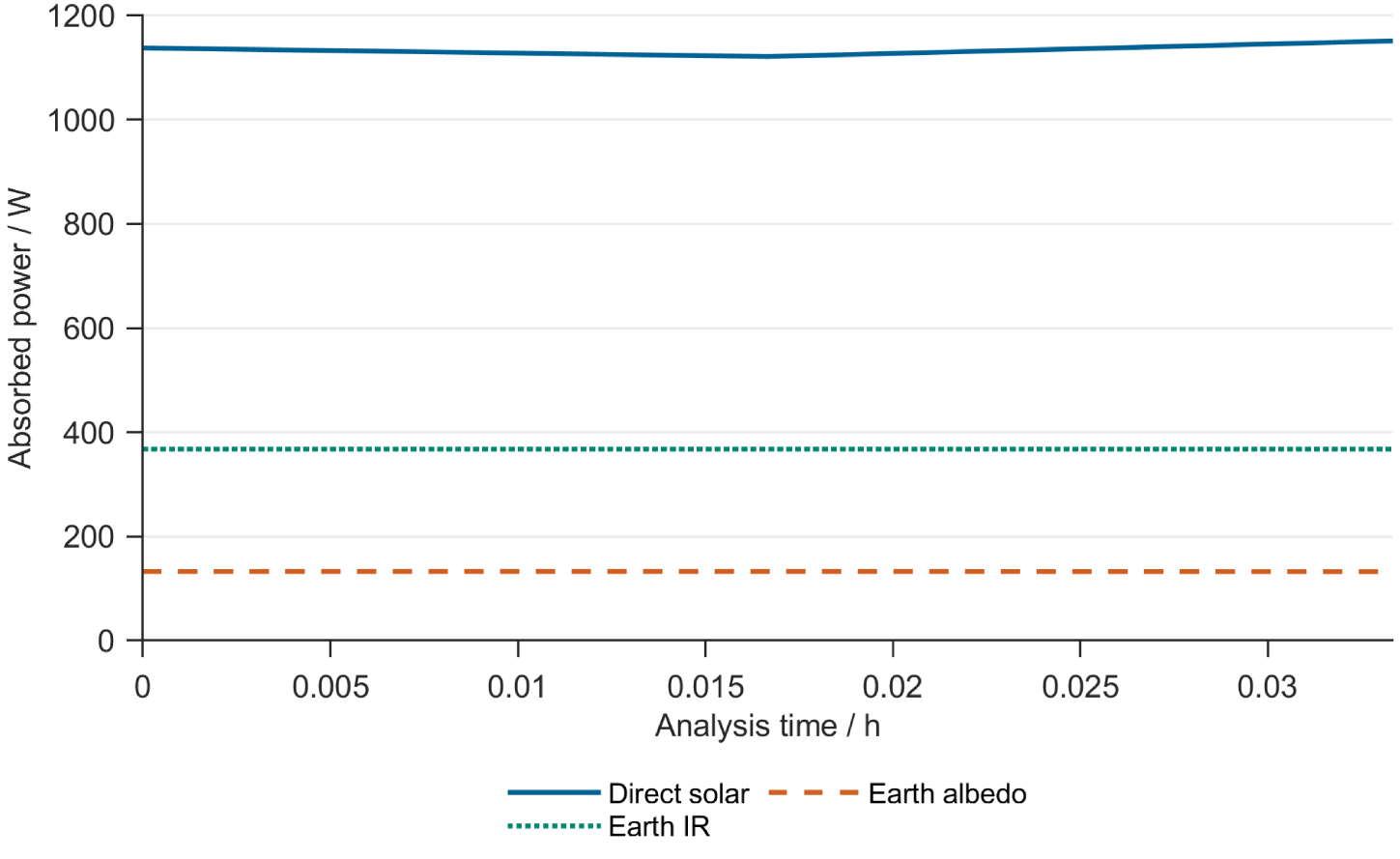
Environmental heat inputs

Time-mean absorbed power: solar 1132.852 W; albedo 132.917 W; Earth IR 367.422 W.

Curves sum absorbed power over the network, including area and absorption properties.

Solar absorbed power 1121.198 to 1151.274 W; Earth IR 367.422 to 367.422 W.

Time-integrated external heat shares are 69.4% direct solar, 8.1% albedo and 22.5% Earth infrared. direct solar contributes most to external absorption; these are heat-input shares, not temperature-response attribution.



Nodal temperatures and extrema

Window maximum 298.596 K (25.45 deg C, node 6); minimum 287.302 K (14.15 deg C, node 3).

Largest full peak-to-peak range 5.848 K (node 3), distinct from a temperature P95-P5 difference.

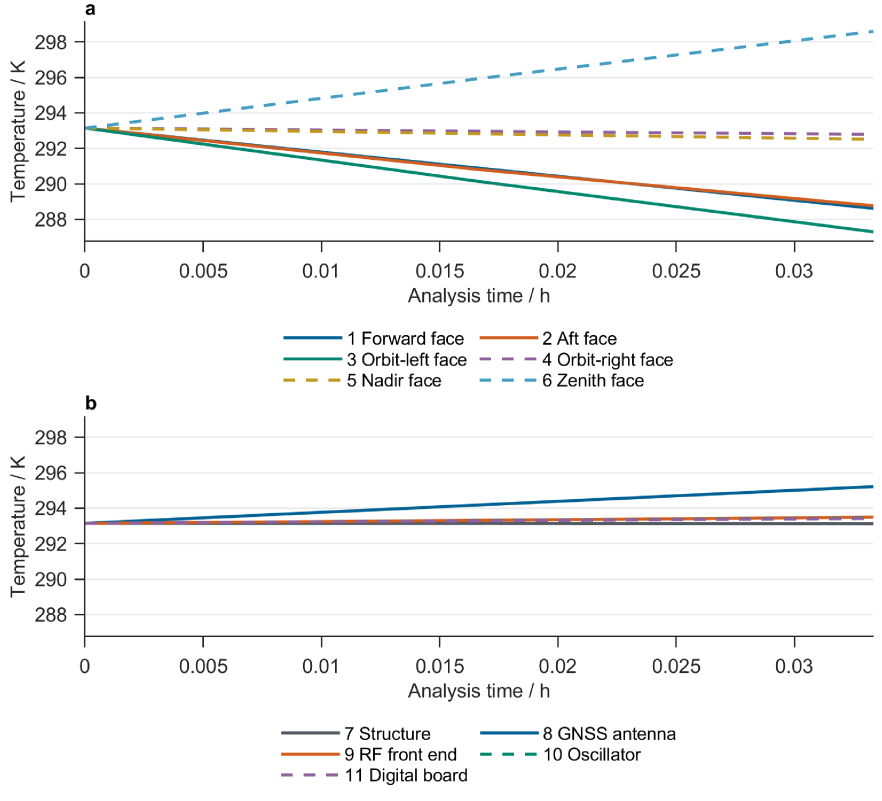
Hottest node: Zenith face; coldest node: Orbit-left face.

11/11 nodes plotted, selected by full temperature range; all are tabulated.

Power-to-temperature peak time lag not calculated: no cycle covers at least 90% of an orbit with at least 8 samples.

Temperature level and variation describe different demands: the maximum is at Zenith face (298.60 K), the minimum at Orbit-left face (287.30 K), and the largest full range at Orbit-left face (5.85 K).

GNSS RF front end has a full range of 0.343 K, 5.9% of the largest nodal range. Assess response-node stability directly rather than substituting exterior or network-wide extremes.

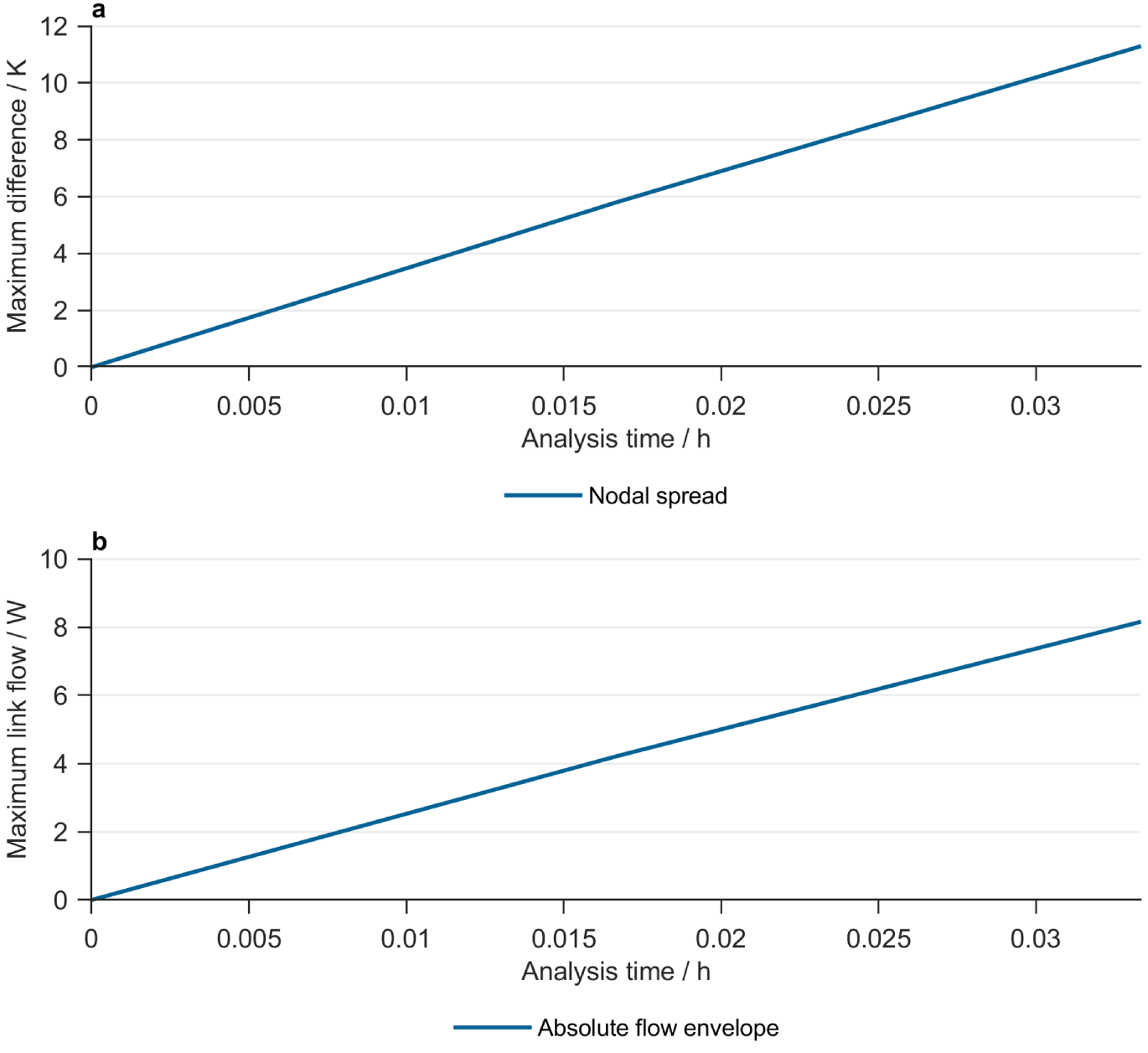


Temperature differences and link heat flow

Maximum simultaneous nodal spread 11.294 K; maximum absolute single-link flow 8.157 W.

Panels show epoch-wise temperature spread and the largest absolute link heat flow.

The envelope takes the largest value across 15 conductive links at each epoch.



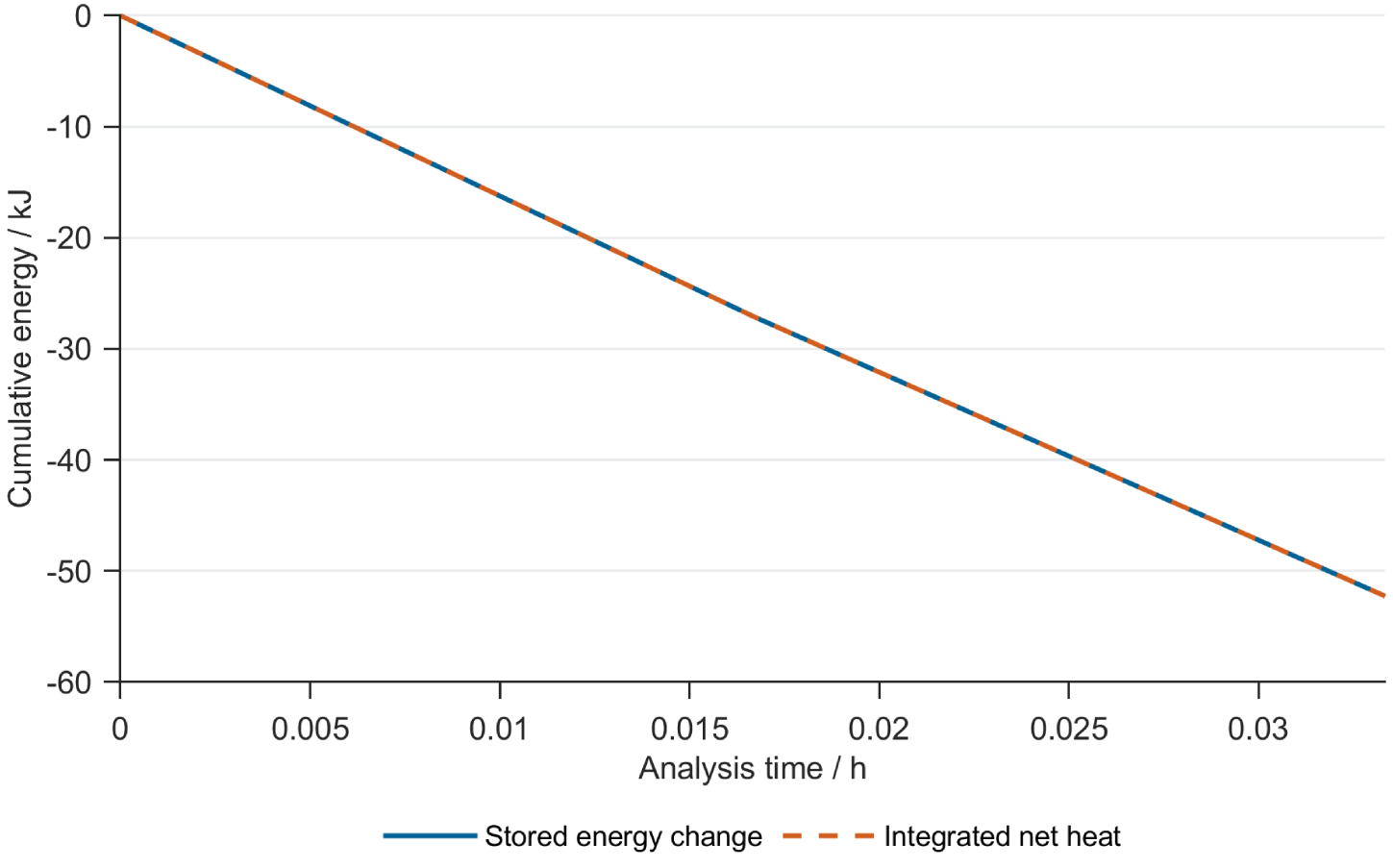
Stored energy and energy balance

After rebasing both at the analysis start, maximum cumulative closure is 4.7294e-10 J.

Stored energy is the sum of capacity times temperature change, relative to analysis start.

Balance: stored energy = external heat + internal heat - radiation + boundary heat.

End-of-window stored-energy change: -52.257 kJ.



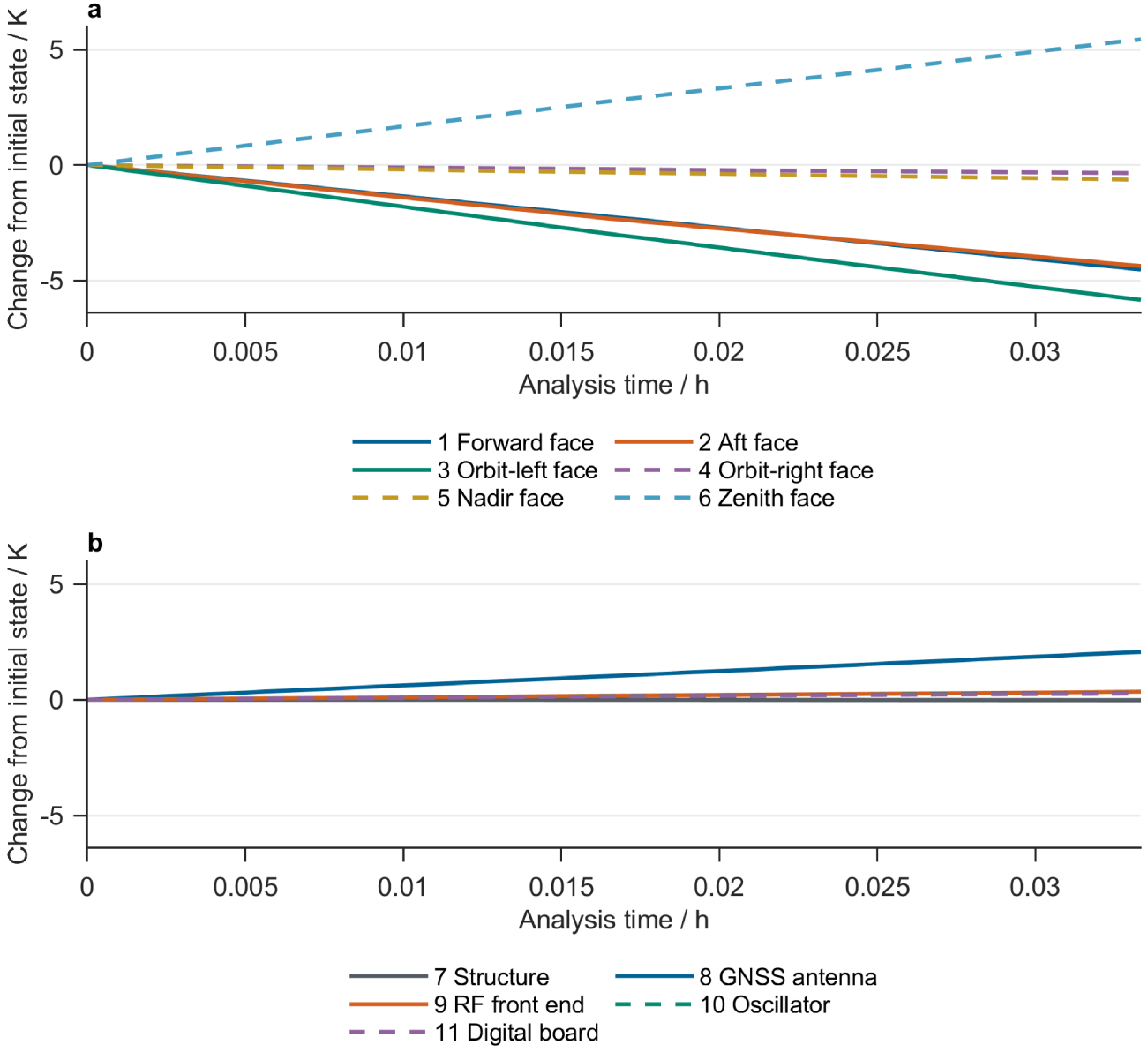
Initial state and temperature drift

Fixed warmup: 0 orbits. Periodic-state convergence was not tested.

Largest absolute end-minus-start temperature change 5.848 K.

Endpoints are 0.021 orbits apart; curves are relative to analysis-start temperatures.

11/11 nodes plotted, selected by full temperature range; all are tabulated.



All-node statistics

Temperature: K; capacity: J/K; configured power: W. Means are time-weighted.

1: Forward face

Min 288.624 | Mean 290.888 | Max 293.150 | Full range 4.526

Capacity 6200 | Internal power 0 | End minus start -4.526 K

2: Aft face

Min 288.773 | Mean 290.885 | Max 293.150 | Full range 4.377

Capacity 6200 | Internal power 0 | End minus start -4.377 K

3: Orbit-left face

Min 287.302 | Mean 290.182 | Max 293.150 | Full range 5.848

Capacity 7000 | Internal power 0 | End minus start -5.848 K

4: Orbit-right face

Min 292.794 | Mean 292.969 | Max 293.150 | Full range 0.356

Capacity 7000 | Internal power 0 | End minus start -0.356 K

5: Nadir face

Min 292.515 | Mean 292.829 | Max 293.150 | Full range 0.635

Capacity 8500 | Internal power 0 | End minus start -0.635 K

6: Zenith face

Min 293.150 | Mean 295.906 | Max 298.596 | Full range 5.446

Capacity 8000 | Internal power 0 | End minus start 5.446 K

All-node statistics (continued)

7: Spacecraft structure

Min 293.129 | Mean 293.143 | Max 293.150 | Full range 0.021

Capacity 26000 | Internal power 2 | End minus start -0.021 K

8: GNSS antenna

Min 293.150 | Mean 294.183 | Max 295.218 | Full range 2.068

Capacity 3200 | Internal power 0.4 | End minus start 2.068 K

9: GNSS RF front end

Min 293.150 | Mean 293.320 | Max 293.493 | Full range 0.343

Capacity 1800 | Internal power 5 | End minus start 0.343 K

10: GNSS oscillator

Min 293.150 | Mean 293.298 | Max 293.445 | Full range 0.295

Capacity 320 | Internal power 0.8 | End minus start 0.295 K

11: Receiver digital board

Min 293.150 | Mean 293.288 | Max 293.425 | Full range 0.275

Capacity 5200 | Internal power 12 | End minus start 0.275 K

Inputs and result details

Scenario name: sample_report_120s

Duration: 120 s

Time step: 60 s

Fixed warmup setting (non-convergence mode): 0 orbit

Node count: 11 node

Conductance range: [0, 1.4] W/K

Internal power range: [0, 12] W

Start epoch UTC: 2024-03-20 12:00:00

Altitude: 600000 m

Inclination: 97.6 deg

Attitude mode: nadir

Trajectory coordinate system: Geocentric equatorial axes (built-in approximate orbit model)

Telemetry comparison: Not run (no telemetry data imported for this task)

Computation checks and audit record

Numerical and configuration checks

Finite scenario temperatures: Pass

Telemetry comparison: Not run (no telemetry data imported for this task)

Assessment scope

This report evaluates the configured thermal simulation and numerical checks. Device temperature-limit compliance is outside this assessment.

Audit files

MAT results, metric CSV, Markdown, manifest, and figures accompany this PDF.

Software version: 1.5

Task conclusions

1. Prioritize the high-temperature level at Zenith face, low-temperature level at Orbit-left face, and variation at Orbit-left face in design review: extrema 298.60/287.30 K and largest range 5.85 K.

2. Prioritize direct solar in environmental-input review: it supplies 69.4% of absorbed external heat. Interpret temperature response together with the heat-transfer paths and orbital phase.

3. The response component GNSS RF front end spans 293.150 to 293.493 K, with a full range of 0.343 K. Use these component values separately from spacecraft extrema.

4. Prioritize the GNSS antenna / GNSS RF front end contact in conductance review: mean net transfer is 0.276 W. This ranking uses observed model heat flow, not a conductance-perturbation experiment.